METRIC

MIL-C-63982A (AR)
AMENDMENT 3
13 August 1998
SUPERSEDING
AMENDMENT 2
16 April 1998

MILITARY SPECIFICATION

CARTRIDGE, 30MM HIGH EXPLOSIVE DUAL PURPOSE, M789

MIL-C-63982 was inactivated after December 6, 1995 for new design.

This amendment forms a part of MIL-C-63982A (AR), dated 26 August 1992, and is approved for use by the US Army Armament Research, Development and Engineering Center, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 2

2.1.2 (DRAWINGS): Delete "9390627 - Barrel, Test, 30mm (ADEN Barrel)"

PAGE 4

3.8: Delete second sentence and substitute "The standard deviation shall not exceed 15 m/s at temperatures of 21^{0}C $\pm$ 5^{0}C and 71^{0}C $\pm$ 5^{0}C. The standard deviation shall not exceed 20 m/s at a temperature of -54^{0}C $\pm$ 5^{0}C.

3.9: Delete in its entirety and substitute the following:

"3.9 Case mouth pressure. In the M230 test barrel (DWG 9390748) the mean peak pressure shall not exceed 335 MPa at 21^{0}C $\pm$ 5^{0}C. The mean peak pressure shall not exceed 430 MPa at temperatures over the range from 54^{0}C to 71^{0}C. Following the exposure to environmental conditions of paragraph 3.10, the mean peak pressure shall not increase nor decrease from the mean peak pressure at 21^{0}C $\pm$ 5^{0}C before exposure by more than 60 MPa."

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PAGE 5

3.11: Delete test barrel DWG. number and Substitute "9390748”

3.15: Delete in its entirety and substitute the following:

“3.15 Normal impact function. When fired against 1.6 mm (nominal thickness) 2024-T3 aluminum alloy sheet, the projectile upon impact shall produce an average hole of not less than seven (7) inches in diameter, exclusive of petals, and show evidence of shrapnel on the forward face or remnants of the aluminum sheet."

PAGE 8

Table I, under Charged Body Assembly, change:

From:	Examination for defects		3.1.2	4.4.2.6
	Static Penetration		3.14	4.5.14
To:	Examination for defects	32	3.1.2	4.4.2.6
	Static Penetration 1/	75	3.14	4.5.14

Table I, add footnote 1/:

Add: 1/ Static penetration testing may be performed in lieu of Dynamic penetration. However, Dynamic penetration shall be performed if the static penetration requirements of 3.14.2 are not met.

PAGE 9

Table I, under Cartridge and Components:

Delete:

"Projectile, 30mm, HEDP, M789 (DWG. 11825955)				
Examination for defects	32		3.1.2	4.4.2.9

And substitute:

“Projectile, 30mm, HEDP, M789 (DWG. 11825955)				
Examination for defects	32		3.1.2	4.4.2.9
Fuze Torque	32	ACC-0 REJ-1	3.5	4.5.16”

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PAGE 11

Table I, note 3/, add:

From: 3/ Fired at 21C.

To: 3/ Fired at 21C. Static penetration testing is not required prior to Dynamic penetration.

PAGE 12

4.4.1: Delete in its entirety and substitute the following:

"4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. Unless otherwise specified, component parts shall be homogeneous and of a quantity convenient to the contractor except that each cartridge lot shall contain:

a. Cartridge cases (DWG 11825902) from one unchanged process, from one lot interfix number and one manufacturer.

b. Projectile metal parts (DWG 9349963) from one unchanged process, from one lot interfix number with not more than two (2) lot numbers from one manufacturer.

c. Primers (DWG 11825910) from one unchanged process, from one lot interfix number with not more than two (2) lot numbers from one manufacturer.

d. Propellant (DWG 11825946) from one lot number from one manufacturer.

e. Fuzes (DWG 11834312) from one unchanged process, from one lot interfix number and one manufacturer."

* 4.5.2: Change

From: 4.5.2 Examination and Tests.

To: 4.4.2 Examination and Tests.

PAGE 14

* 4.4.2.1: Delete notes at bottom of page.

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PAGE 15

* 4.4.2.1: Change notes:

From: NOTES: ** Test may be conducted using the same sample cartridges.

TO: NOTES: * Test may be conducted using the same sample cartridges.
** Test may be conducted using the same sample cartridge.

* 4.4.2.1: Change:

From: /Major 118/Function & Casualty/Table VII 0-1

To: /Major 118/Function & Casualty,/Table IV, VI/3.13/4.5.5
M230 Autogun 0-1

Under Ballistic Performance (21C), add:

Dynamic Penetration 8/	25, 4-11 50, 10-11	3.14.1	4.5.14.1
Static Penetration 8/	75	3.14.2	4.5.14.2

PAGE 21

* 4.4.2.6: Delete Major 104

Delete: /Major/ Weight / Level II / 3.1.1 / Gage
104

PAGE 22

4.4.2.7, Major 101: Delete "Visual/Weight 1/" and substitute "Weight/Gage 1/".

PAGE 24

* 4.4.2.9: Change:

From: Paragraph 4.4.2.9 Title: Projectile, 30mm, HEDP, M789
Sheet: 1 of 1
Drawing Number: 11825954
Next Higher Assembly:

TO: Paragraph 4.4.2.9 Title: Projectile, 30mm, HEDP, M789
Sheet: 1 of 1
Drawing Number: 11825955
Next Higher Assembly: 11825954

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4.4.2.9: Add:

"104 Presence of thread lock Level II 3.1.2 Gage 1/"

To note block add: "1/ Presence of thread lock shall be checked using automated methods prior to assembly of fuze to charged body."

PAGE 28

Add note 8/:

Add: “8/ Static penetration testing may be performed in lieu of Dynamic penetration. However, Dynamic penetration shall be performed if the static penetration requirements of 4.5.14.2 are not met. Static penetration is not required prior to Dynamic penetration. Failure of 11 or more samples shall cause rejection of the lot. If more than 4, but less than 11 fail, a second sample of 25 cartridges shall be tested. The lot shall be rejected if 11 or more cartridges of the combined sample of 50 fail.”

PAGE 33

* Table VII: Delete in its entirety.

PAGE 34

4.5.1: Delete test barrel DWG. number and substitute "9390748".

PAGE 35

* 4.5.4: Delete second to last sentence and substitute the following:

"The time shall comply with requirements of 3.12."

4.5.4: Delete last sentence and substitute the following:

"Any corrected action time of 1.92 milliseconds or less shall be disregarded as instrumentation error."

* 4.5.5: Delete in its entirety and substitute the following:

"4.5.5 Function and casualty. These tests shall be conducted, using the test methods prescribed in AMCR-715-505, Volume 8, Chapter 1, Section 3 as a guide. Witness panels for structural integrity shall be placed approximately 5 meters from the muzzle. A failure is characterized by observance of any of the firing defects listed in Table VI. Test rounds, when required, shall be fired in the M230 Gun. Test rounds shall be conditioned at 21^{0}C. Test results shall conform to the requirements of 3.13."

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PAGE 35 (conintued)

* 4.5.6: Delete in its entirety and substitute the following:

"4.5.6 Projectile structural integrity. Projectile structural integrity shall be verified in conjunction with the ballistic tests of 4.5.2, 4.5.3 and 4.5.4. A witness board/card shall be located approximately 5 meters from the gun muzzle and a witness box shall be located at the muzzle and extend at least one meter forward from the muzzle. The witness box shall be open at each end and be lined with 1/2 inch thick fiberboard. Loss of projectile structural integrity is defined as projectile fracture, break-up, rotating band failure as defined in 6.13, or loss of parts as evidenced by microflash photograph and/or hole(s) in the witness board or recovery of fragments."

4.5.7: Delete in its entirety and substitute the following:

"4.5.7 Environmental tests. Unless otherwise specified, performance after environmental exposure, as specified in the applicable test procedure, shall consist of conducting the tests of 4.5.2, 4.5.3, 4.5.4 and 4.5.14.1, except all test rounds shall be conditioned at 21^{0}C ± 5^{0}C prior to firing. Seventy-five percent (75%) of each environmental test sample shall be designated for test 4.5.2, 4.5.3 and 4.5.4 and the balance (25%) designated for test 4.5.14.1."

PAGE 36

Figure 1: Delete in its entirety and substitute with attached Figure 1, Vertical vibration test schedule.

Add attached Table VIII, Vertical vibration test schedule.

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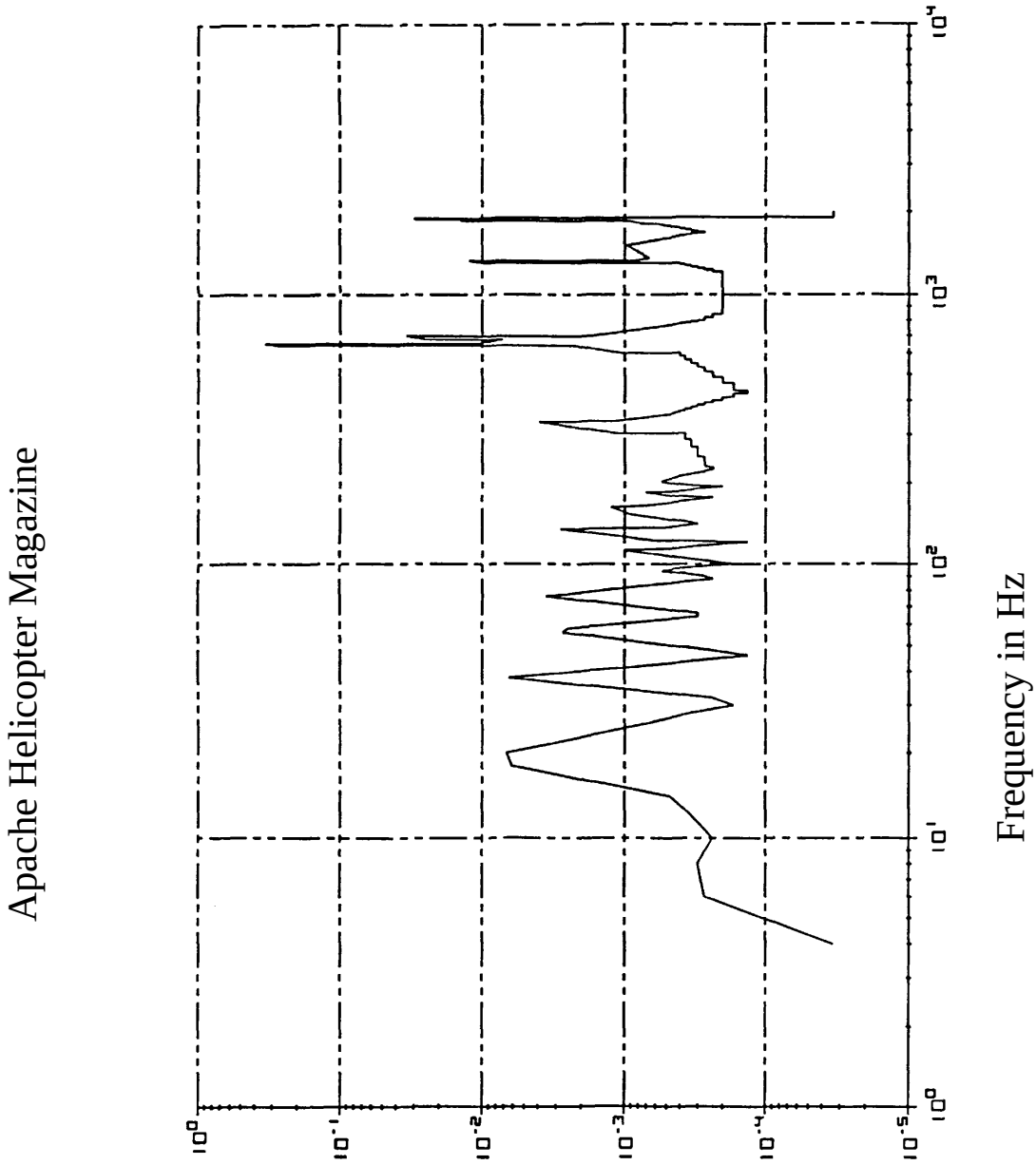


FIGURE 1: Vertical vibration test schedule

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TABLE VIII. Vertical vibration test schedule.

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AIR COMBAT SCHEDULE EF=1.0 VERTICAL TEST TIME = 127 MINUTES

50 NUMBER OF BREAK POINTS

5.00	0.0002000	FREQUENCY AND AMPLITUDE
7.00	0.0003800	FREQUENCY AND AMPLITUDE
10.00	0.0002400	FREQUENCY AND AMPLITUDE
14.00	0.0005000	FREQUENCY AND AMPLITUDE
19.00	0.0106100	FREQUENCY AND AMPLITUDE
31.00	0.0001400	FREQUENCY AND AMPLITUDE
38.00	0.0064400	FREQUENCY AND AMPLITUDE
46.00	0.0001400	FREQUENCY AND AMPLITUDE
57.00	0.0035600	FREQUENCY AND AMPLITUDE
65.00	0.0002400	FREQUENCY AND AMPLITUDE
76.00	0.0035600	FREQUENCY AND AMPLITUDE
89.00	0.0002200	FREQUENCY AND AMPLITUDE
95.00	0.0006500	FREQUENCY AND AMPLITUDE
100.00	0.0002000	FREQUENCY AND AMPLITUDE
114.00	0.0013100	FREQUENCY AND AMPLITUDE
121.00	0.0001400	FREQUENCY AND AMPLITUDE
135.00	0.0031400	FREQUENCY AND AMPLITUDE
142.00	0.0002400	FREQUENCY AND AMPLITUDE
152.00	0.0009300	FREQUENCY AND AMPLITUDE
164.00	0.0013300	FREQUENCY AND AMPLITUDE
169.00	0.0004400	FREQUENCY AND AMPLITUDE
178.00	0.0002100	FREQUENCY AND AMPLITUDE
186.00	0.0008600	FREQUENCY AND AMPLITUDE
196.00	0.0001800	FREQUENCY AND AMPLITUDE
203.00	0.0005900	FREQUENCY AND AMPLITUDE
223.00	0.0002600	FREQUENCY AND AMPLITUDE
304.00	0.0004000	FREQUENCY AND AMPLITUDE
335.00	0.0041600	FREQUENCY AND AMPLITUDE
357.00	0.0004800	FREQUENCY AND AMPLITUDE
428.00	0.0001600	FREQUENCY AND AMPLITUDE
602.00	0.0004400	FREQUENCY AND AMPLITUDE
642.00	0.0023100	FREQUENCY AND AMPLITUDE
655.00	0.3429600	FREQUENCY AND AMPLITUDE
677.00	0.0070400	FREQUENCY AND AMPLITUDE
698.00	0.0348600	FREQUENCY AND AMPLITUDE
795.00	0.0003100	FREQUENCY AND AMPLITUDE
865.00	0.0002100	FREQUENCY AND AMPLITUDE
1191.00	0.0002100	FREQUENCY AND AMPLITUDE
1297.00	0.0004500	FREQUENCY AND AMPLITUDE
1325.00	0.0124000	FREQUENCY AND AMPLITUDE
1350.00	0.0006900	FREQUENCY AND AMPLITUDE
1506.00	0.0009800	FREQUENCY AND AMPLITUDE
1675.00	0.0003000	FREQUENCY AND AMPLITUDE
1805.00	0.0005900	FREQUENCY AND AMPLITUDE
1855.00	0.0009700	FREQUENCY AND AMPLITUDE
1866.00	0.0142400	FREQUENCY AND AMPLITUDE
1873.00	0.0069000	FREQUENCY AND AMPLITUDE
1880.00	0.0148300	FREQUENCY AND AMPLITUDE
1890.00	0.0301400	FREQUENCY AND AMPLITUDE
1904.00	0.0099700	FREQUENCY AND AMPLITUDE

RMS VALUE =2.65

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* Figure 2: Delete in its entirety and substitute with attached Figure 2, Transverse vibration test schedule.

* Add attached Table IX, Transverse vibration test schedule.

* Add attached Figure 3, Longitudinal vibration test schedule.

* Add attached Table X Longitudinal vibration test schedule.

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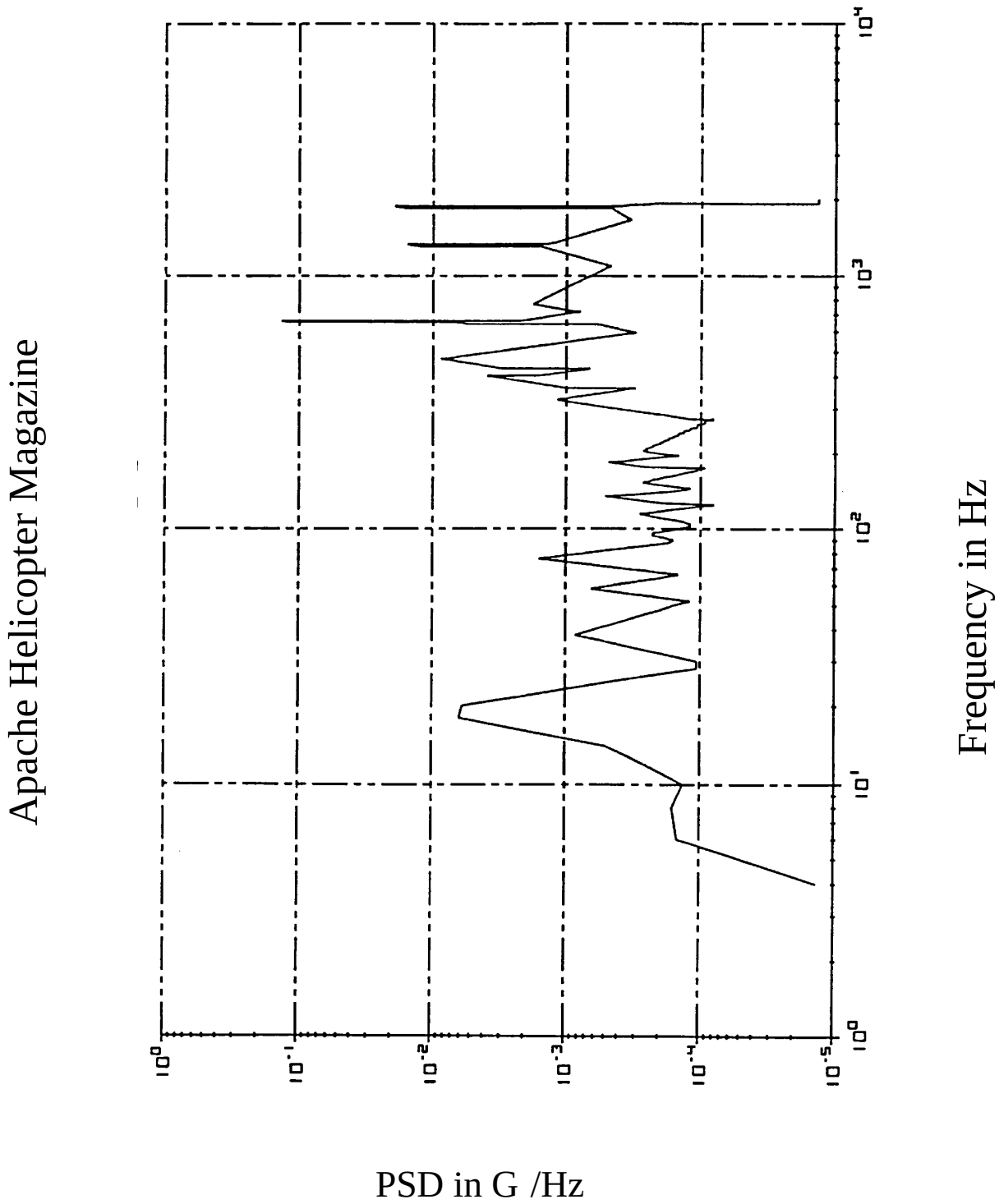


FIGURE 2: Transverse vibration test schedule

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TABLE IX. Transverse vibration test schedule.

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AIR COMBAT SCHEDULE EF=1.0 TRANSVERSE TEST TIME = 127 MINUTES

45	NUMBER OF BREAK POINTS	
5.00	0.0001300	FREQUENCY AND AMPLITUDE
7.00	0.0001800	FREQUENCY AND AMPLITUDE
10.00	0.0001400	FREQUENCY AND AMPLITUDE
14.00	0.0005000	FREQUENCY AND AMPLITUDE
19.00	0.0106100	FREQUENCY AND AMPLITUDE
29.00	0.0000800	FREQUENCY AND AMPLITUDE
38.00	0.0008500	FREQUENCY AND AMPLITUDE
52.00	0.0001300	FREQUENCY AND AMPLITUDE
58.00	0.0006500	FREQUENCY AND AMPLITUDE
66.00	0.0001500	FREQUENCY AND AMPLITUDE
76.00	0.0015800	FREQUENCY AND AMPLITUDE
89.00	0.0001500	FREQUENCY AND AMPLITUDE
95.00	0.0002600	FREQUENCY AND AMPLITUDE,
103.00	0.0001100	FREQUENCY AND AMPLITUDE
114.00	0.0002900	FREQUENCY AND AMPLITUDE
125.00	0.0000800	FREQUENCY AND AMPLITUDE
135.00	0.0005700	FREQUENCY AND AMPLITUDE
144.00	0.0001100	FREQUENCY AND AMPLITUDE
152.00	0.0002800	FREQUENCY AND AMPLITUDE
176.00	0.0000900	FREQUENCY AND AMPLITUDE
186.00	0.0005700	FREQUENCY AND AMPLITUDE
196.00	0.0001600	FREQUENCY AND AMPLITUDE
205.00	0.0002800	FREQUENCY AND AMPLITUDE
273.00	0.0000900	FREQUENCY AND AMPLITUDE
327.00	0.0012100	FREQUENCY AND AMPLITUDE
364.00	0.0002900	FREQUENCY AND AMPLITUDE
405.00	0.0039700	FREQUENCY AND AMPLITUDE
432.00	0.0006300	FREQUENCY AND AMPLITUDE
471.00	0.0089300	FREQUENCY AND AMPLITUDE
595.00	0.0003100	FREQUENCY AND AMPLITUDE
641.00	0.0005800	FREQUENCY AND AMPLITUDE
656.00	0.0069500	FREQUENCY AND AMPLITUDE
662.00	0.1388600	FREQUENCY AND AMPLITUDE
676.00	0.0016900	FREQUENCY AND AMPLITUDE
721.00	0.0007900	FREQUENCY AND AMPLITUDE
769.00	0.0017700	FREQUENCY AND AMPLITUDE
1093.00	0.0004800	FREQUENCY AND AMPLITUDE
1297.00	0.0015600	FREQUENCY AND AMPLITUDE
1325.00	0.0156600	FREQUENCY AND AMPLITUDE
1339.00	0.0012300	FREQUENCY AND AMPLITUDE
1658.00	0.0003400	FREQUENCY AND AMPLITUDE
1844.00	0.0004700	FREQUENCY AND AMPLITUDE
1884.00	0.0194800	FREQUENCY AND AMPLITUDE
1904.00	0.0003400	FREQUENCY AND AMPLITUDE
1925.00	0.0002300	FREQUENCY AND AMPLITUDE
RMS VALUE =	1.92	

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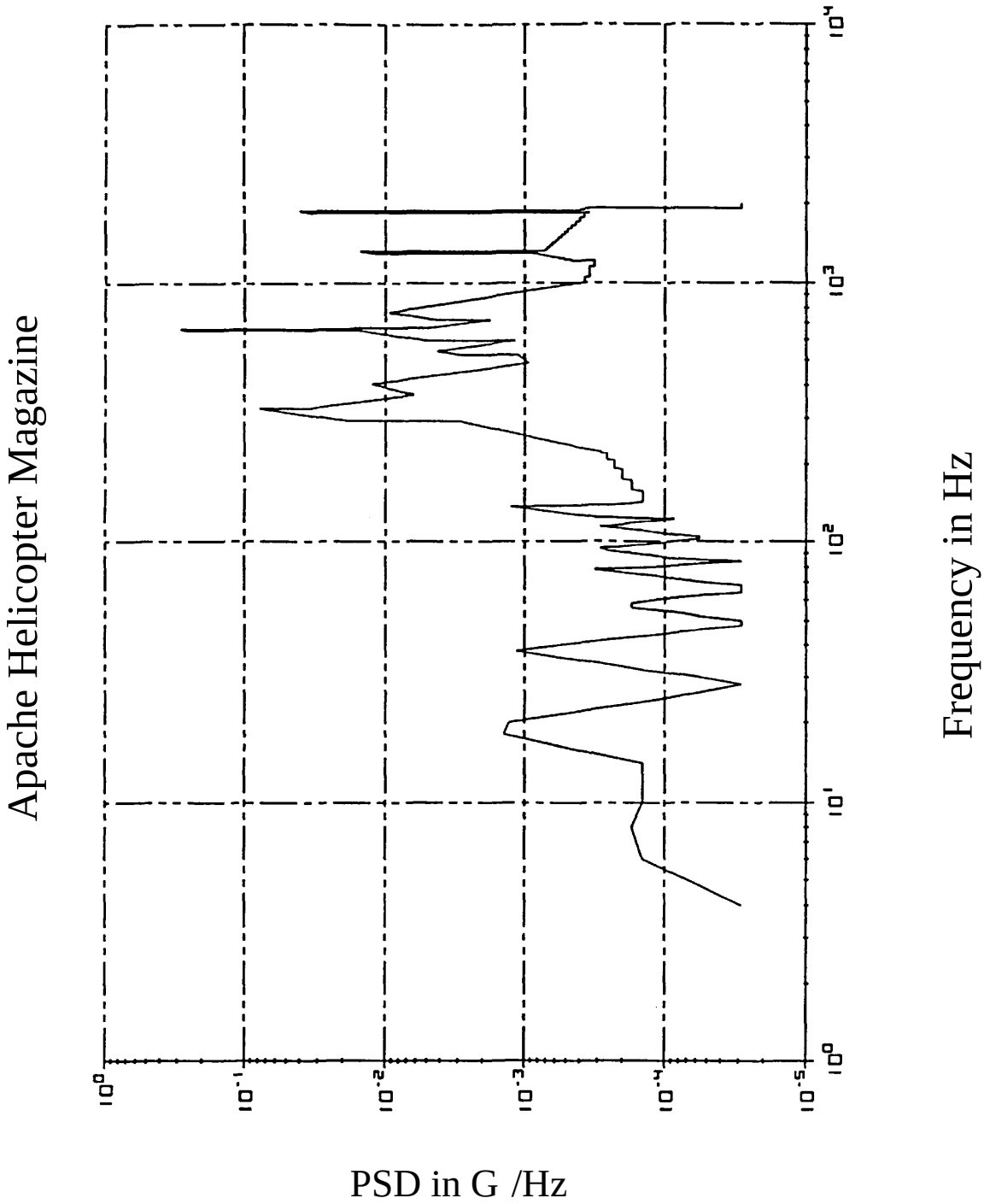


FIGURE 3 : Longitudinal vibration test schedule

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TABLE X. Longitudinal vibration test schedule.

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AIR COMBAT SCHEDULE EF=l.0 VERTICAL TEST TIME = 127 MINUTES

39 NUMBER OF BREAK POINTS

5.00	0.0001400	FREQUENCY AND AMPLITUDE
7.00	0.0001800	FREQUENCY AND AMPLITUDE
14.00	0.0001600	FREQUENCY AND AMPLITUDE
19.00	0.0023000	FREQUENCY AND AMPLITUDE
28.00	0.0000300	FREQUENCY AND AMPLITUDE
38.00	0.0011600	FREQUENCY AND AMPLITUDE
49.00	0.0000300	FREQUENCY AND AMPLITUDE
57.00	0.0002400	FREQUENCY AND AMPLITUDE
67.00	0.0000300	FREQUENCY AND AMPLITUDE
80.00	0.0004800	FREQUENCY AND AMPLITUDE
86.00	0.0000300	FREQUENCY AND AMPLITUDE
94.00	0.0003800	FREQUENCY AND AMPLITUDE
103.00	0.0000500	FREQUENCY AND AMPLITUDE
114.00	0.0002900	FREQUENCY AND AMPLITUDE
124.00	0.0000900	FREQUENCY AND AMPLITUDE
137.00	0.0014400	FREQUENCY AND AMPLITUDE
142.00	0.0001500	FREQUENCY AND AMPLITUDE
223.00	0.0002900	FREQUENCY AND AMPLITUDE
294.00	0.0030500	FREQUENCY AND AMPLITUDE
327.00	0.0804700	FREQUENCY AND AMPLITUDE
368.00	0.0062400	FREQUENCY AND AMPLITUDE
405.00	0.0125900	FREQUENCY AND AMPLITUDE
486.00	0.0009600	FREQUENCY AND AMPLITUDE
524.00	0.0011400	FREQUENCY AND AMPLITUDE
541.00	0.0042900	FREQUENCY AND AMPLITUDE
595.00	0.0011600	FREQUENCY AND AMPLITUDE
655.00	0.0159100	FREQUENCY AND AMPLITUDE
662.00	0.2980700	FREQUENCY AND AMPLITUDE
669.00	0.0166700	FREQUENCY AND AMPLITUDE
721.00	0.0017700	FREQUENCY AND AMPLITUDE
769.00	0.0093600	FREQUENCY AND AMPLITUDE
1004.00	0.0003900	FREQUENCY AND AMPLITUDE
1216.00	0.0003300	FREQUENCY AND AMPLITUDE
1297.00	0.0008800	FREQUENCY AND AMPLITUDE
1325.00	0.0151800	FREQUENCY AND AMPLITUDE
1339.00	0.0007300	FREQUENCY AND AMPLITUDE
1844.00	0.0003700	FREQUENCY AND AMPLITUDE
1884.00	0.0405300	FREQUENCY AND AMPLITUDE
1925.00	0.0003400	FREQUENCY AND AMPLITUDE

RMS VALUE = 2.98

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PAGE 38

4.5.7.3: Delete in its entirety and substitute the following:

"4.5.7.3 Aircraft vibration. The ammunition loaded in an Apache magazine shall be vibrated according to Figures 1, 2, & 3 and Tables VIII, IX, & X. The magazine shall be mounted to a test fixture in three orientations: vertical, transverse, and longitudinal. The test duration shall be 127 minutes in each orientation and at each temperature of -40^{0}C and 65^{0}C. After vibration the ammunition shall be subjected to the tests in 4.5.9."

PAGE 40

4.5.15: Delete in its entirety and substitute the following:

"4.5.15 Normal impact function. The target shall be placed 200 meters +/- 20 meters from the muzzle. Failure of any projectile to function high order detonation and produce a hole as defined in paragraph 3.15 shall be considered a defect. Failure of 11 or more projectiles shall be cause for rejection of the lot. If 3 to 10 failures occur, testing will continue in accordance with the following double sampling plan:

nl=90	cl=2	rl=1l
nt=270	ct=l0	rt=ll

If two (2) or less failures occur during the initial test sample, the lot will be accepted."

PAGE 42

* 5.1.2.3: Change:

From: 1291089

TO: 12910289

PAGE 43

5.3: After "...charge weights, projectile" add ",fuze and lot identification."

PAGE 45

6.14: Delete "SMCAR-QAF-I(D)" and substitute "AMSTA-AR-QAC-C".

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